ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

S. Rupika

192512398

C01-4

EXPERIMENT 4

AIM:

To write program for continuous Charge Distribution in electric field. The different types of charge density are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using MATLAB.

Test case (i) Determination of line, surface and volume charge density for given charge, length and radius

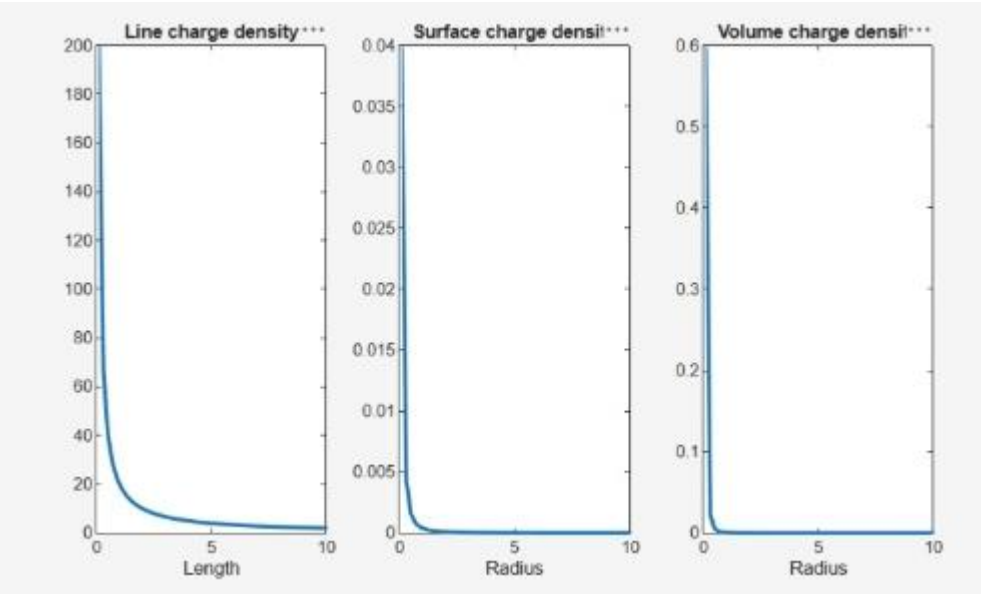
```
clc;
clear;
q=input("Enter the charge value is:");
 l=input("Enter the length value is:");
 % Line charge density rho_l = q/l;
disp('the line charge density is');
disp(rho_l);
 % Surface charge density r=input("Enter the radius value is:");
 rho_s = q/(4*pi*r^2);
disp('the surface charge density is');
disp(rho_s);
% Volume charge density rho_v = q/((4/3)*pi*r^3);
 disp('the volume charge density is');
 disp(rho_v);
% Plot x = linspace(0.1,10,50);
y = rho_l./x; z = rho_s./(x.^2);
 v = rho_v./(x.^3);
 figure; subplot(1,3,1);
 plot(x,y,'LineWidth',2);
title('Line charge density');
xlabel('Length');
subplot(1,3,2);
plot(x,z,'LineWidth',2);
title('Surface charge density');
 xlabel('Radius');
 subplot(1,3,3);
plot(x,v,'LineWidth',2);
title('Volume charge density');
 xlabel('Radius');
```

Output:-

```
 Enter the charge value is: 2e-2
Enter the length value is: 1e-3 t
he line charge density is 20
Enter the radius value is: 2
 the surface charge density is 3.98e-4
```

the volume charge density is 5.97e-4

OUTPUT:



Test case (ii) A long wire of length L = 10 m carries a uniform charge of Q = 50 µC. Calculate the line charge density and plot electric field variation

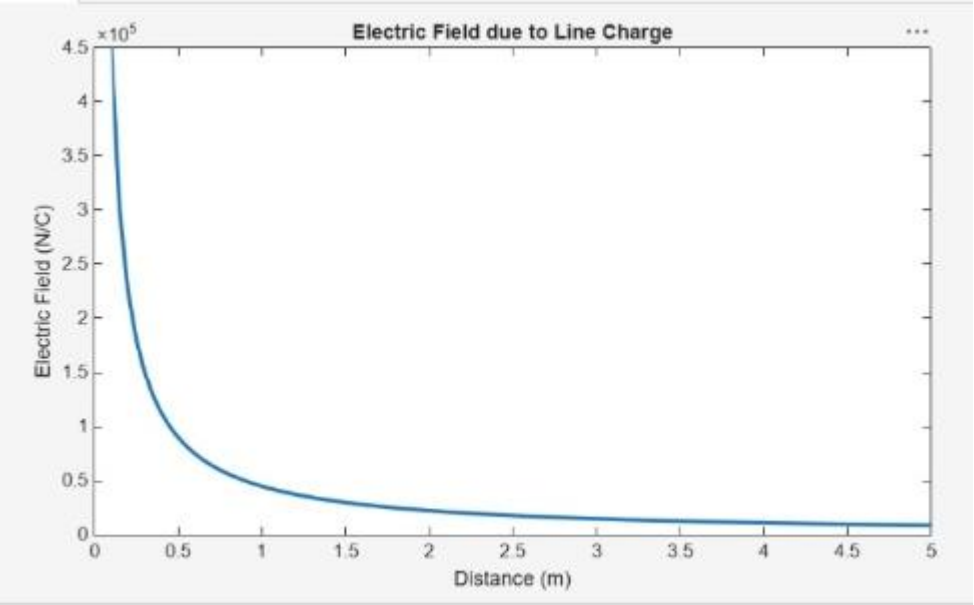
Program:

```
 clc;
L = 10;
 Q = 50e-6;
lambda = Q/L;
disp("Line charge density:");
 disp(lambda);
r = linspace(0.1,5,100);
 k = 9e9;
 E = k*lambda./r;
 plot(r,E,'LineWidth',2);
 xlabel('Distance (m)');
 ylabel('Electric Field (N/C)');
 title('Electric Field due to Line Charge');
```

Output:-

- Line charge density = 5×10^{-6} C/m
- Electric field decreases with distance

OUTPUT:



Test case (iii) A uniformly charged disk of radius R = 0.5 m with total charge Q = 10 μC. Calculate surface charge density and electric field

Program:

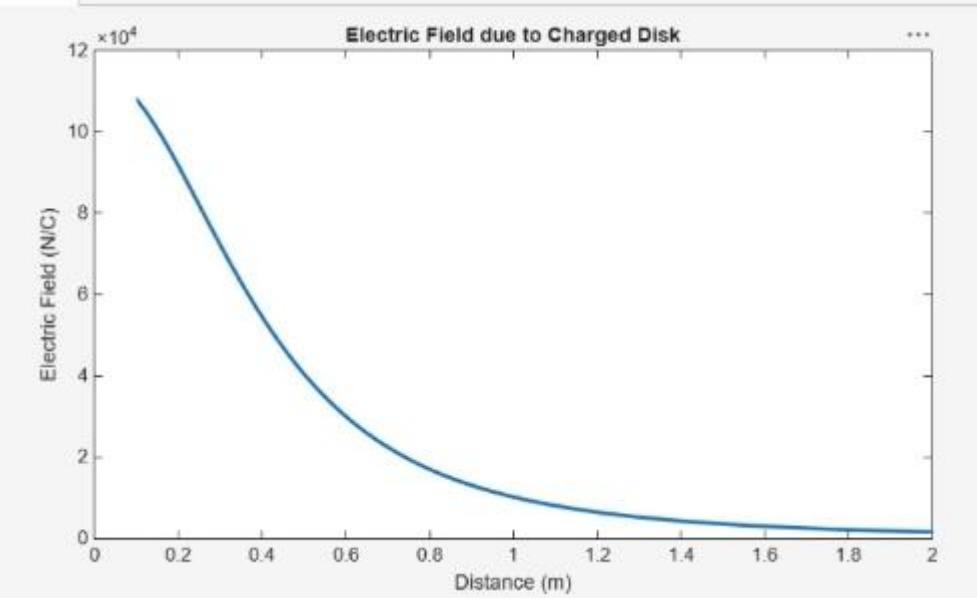
```
clc;
R = 0.5;
Q = 10e-6;
A = pi*R^2;
sigma = Q/A;
disp("Surface charge density:");
disp(sigma);
z = linspace(0.1,2,100);
k = 9e9;
E = (k*sigma*R^2)./(2*(z.^2 + R^2).^(3/2));
plot(z,E,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Electric Field (N/C)');
title('Electric Field due to Charged Disk');
```

Output:-

- Surface charge density = 1.2732e-05

● Electric field decreases non-linearly

OUTPUT:



Test case (iv): A uniformly charged sphere of radius R = 2 m with total charge Q = 20 µC. Calculate volume charge density and electric field inside and outside 3

Program:

```
clc;
R = 2;
 Q = 20e-6;
 V = (4/3)*pi*R^3;
 rho = Q/V;
 disp("Volume charge density:");
 disp(rho);
r = linspace(0.1,3,100);
 epsilon0 = 8.854e-12;
 k = 1/(4*pi*epsilon0);
 E_inside = (rho.*r)/(3*epsilon0);
E_outside = k*Q./(r.^2);
 plot(r,E_inside,'LineWidth',2);
 hold on;
 plot(r,E_outside,'LineWidth',2);
 xlabel('Distance (m)');
 ylabel('Electric Field (N/C)');
 title('Electric Field inside and outside sphere');
legend('Inside','Outside');
```

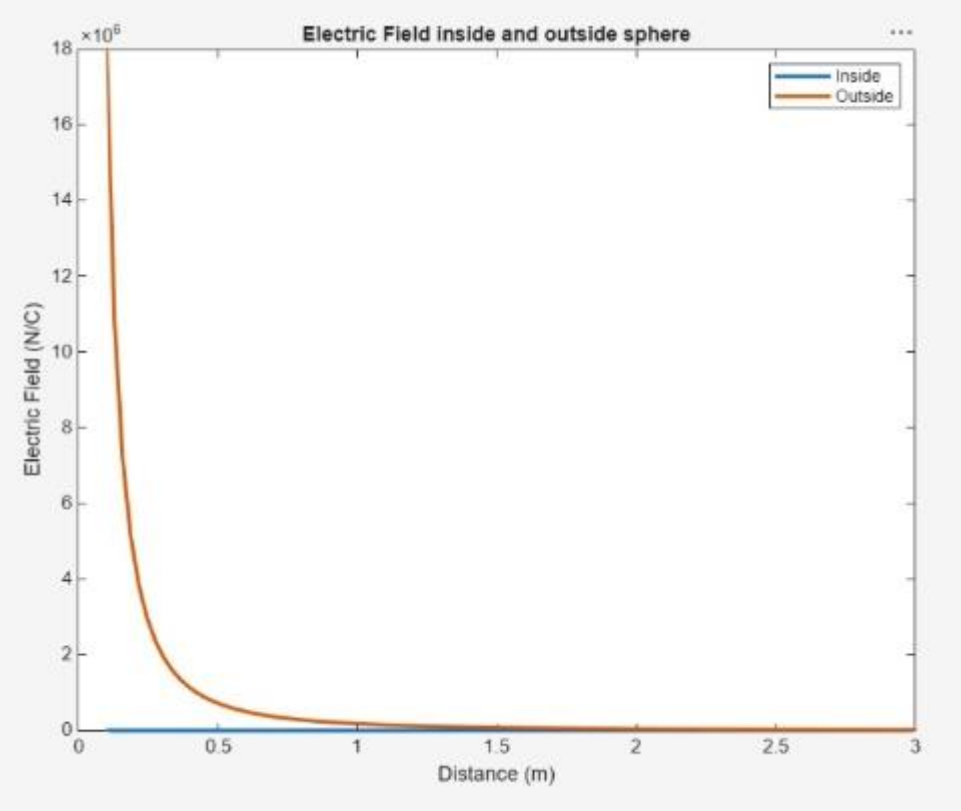
Output:

● Volume charge density: 5.9683e-07

● Inside: linear variation

● Outside: inverse square variation

OUTPUT:


Electric Field inside and outside sphere
×10^6
Inside
Outside
Electric Field (N/C)
Distance (m)
0
2
4
6
8
10
12
14
16
18
0
0.5
1
1.5
2
2.5
3

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage
Identify physical quantities and force models	Identify physical quantities and force models	Correctly identifies charge/current, separation, medium and force direction for both cases.	Identifies most quantities and models correctly.	Needs guidance in selecting the correct relation or direction.	Uses incorrect force models or quantities.	25
Calculate electric and magnetic forces	Calculate electric and magnetic forces	Implements Coulomb-force and parallel-current force calculations correctly with units.	Calculations are mostly correct with minor errors.	Some calculations are correct but require assistance.	Force calculations are invalid or absent.	25
Evaluate force versus separation	Evaluate force versus separation	Correctly varies separation, plots trends and checks theoretical distance dependence.	Trend and plots are correct with minor omissions.	Limited sweep or incomplete trend interpretation.	Does not establish a valid relationship.	30
Objective	**Assessment Criterion**	Excellent – 4	Good – 3	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightage**
Explain and report results	**Explain and report results**	Explains attraction/repulsion and reports assumptions, units and conclusions clearly.	Conclusion is correct with minor documentation gaps.	**Conclusion is incomplete or partly inaccurate.**	**Conclusion is missing or incorrect.**	**20**